

Quantifying Routing-Induced Frequency Dispersion in an Open-Source SKY130 Ring-Oscillator PUF: A Pre-Silicon Study

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Abstract

A ring-oscillator physical unclonable function (RO-PUF) turns manufacturing variation between nominally identical oscillators into a device secret. The physical implementation can add a frequency pattern of its own on top of that variation. This paper quantifies one component of that pattern, the instance-to-instance routing capacitance assigned by the automated OpenLane/OpenROAD flow on the open SKY130 process design kit, using a reduced model: nominal transistor-level models of the verified post-route oscillator topology, loaded with each ring net's total extracted capacitance from the final SPEF. No fabricated devices have been measured yet, and distributed resistance and coupling are not represented.

Under this model, the 16 automatically placed oscillators of Arm A in a coherent build of the current RTL spread 10.5% peak to peak (population SD 2.36%, mean 551.7 MHz). One oscillator with an unusually heavy routing load (24.4 fF against a 11.7 to 17.0 fF range for the rest) contributes much of the extreme range: without it the peak-to-peak figure is 4.6% and the SD 1.25%, while the capacitance-frequency relation is essentially unchanged. Two earlier uncontrolled builds showed dispersions of 8.8% and 5.4% with different per-instance patterns; they are supporting historical observations, not controlled replicates. A linear capacitance fit trained on the earlier 32-oscillator build predicts the current build's 16 frequencies with a mean absolute error of 0.14% and rank correlation 0.997, so the capacitive mechanism transfers across builds even though each pattern is unique to its routing run. The comparison arm, Arm B, uses one hardened oscillator macro sixteen times. A single extraction of that macro gives a 569.5 MHz reference for the shared internal layout, which removes internal-layout variation by construction but does not on its own prove a smaller total spread once the chips are made.

The contribution is a pre-fabrication diagnostic, traceable in the repository, that separates nominal layout bias from the variation a PUF is supposed to use. Whether the layout pattern repeats across dies, reduces uniqueness, or enables a practical prediction attack is left for the silicon phase, not claimed here.

1. Introduction

An RO-PUF compares the frequencies of nominally identical ring oscillators and converts comparisons into response bits [1]. Ideally the useful chip-to-chip differences come from

manufacturing variation. In practice, placement and routing give instances different parasitic loads, which adds a deterministic component to the comparison.

A fixed layout pattern is not automatically a security failure. Random mismatch may be larger, comparisons may cancel shared structure, and response processing may absorb bias. But a layout component that repeats across chips could reduce uniqueness or make some comparisons predictable, and deciding between those outcomes takes measurements from multiple fabricated devices.

This paper reports the pre-silicon part of that investigation. The question is narrow: with transistor parameters held fixed, how much frequency variation does the physical implementation of one open-source RO-PUF design introduce? I contribute nominal post-layout results for two automatically placed oscillator arrays, a capacitance-based explanation for the observed spread, a matched-macro arm that gives every instance the same internal geometry, the scripts to run the same diagnostic before any fabrication, and the specific predictions I plan to test once chips come back.

2. Background and related work

Suh and Devadas introduced the widely used RO-PUF construction [1]. Maiti and Schaumont examined improved ring-oscillator PUF designs and compensation for systematic effects [2], and later FPGA studies mapped spatial variation and placement-dependent behaviour [3, 5]. Other work proposes statistical bias reduction, configurable structures, and placement-aware designs [5-8]. Katzenbeisser et al. evaluated several PUF constructions, including ring oscillators, across 96 ASICs [12].

The literature does not support a simple rule that any systematic structure makes an RO-PUF predictable. Wilde, Hiller, and Pehl found that adjacent-oscillator comparisons reduced exploitable spatial structure in their data, and that the estimated covariance was too small for their predictor to beat the relevant baseline [4]. That is important counterevidence: layout bias has to be judged together with the comparison scheme, the mismatch distribution, and the attacker model.

Open-source ASIC flows allow an experiment that closed flows make awkward: the designer can read the routed netlist and the parasitic extraction before fabrication instead of treating the implementation as opaque. OpenLane/OpenROAD [9] and the open SKY130 PDK [10] provide that setting, and a TinyTapeout RO-PUF project shows the circuit family works in the same ecosystem [11]. What I did not find is the measurement this paper makes: the nominal frequency component tied to instance-dependent routing in one automated ASIC layout, quantified from the flow's own extraction.

3. Design under test

The candidate design, `tt_um_nikodemetrashvili20_ro_puf`, occupies a TinyTapeout 2x2 tile. It holds two 16-oscillator arms and a shared serial measurement core that enables one oscillator at a time and counts its edges over a fixed window.

Each oscillator is a 31-stage ring of SKY130 standard cells: an enable NAND, 30 inverters, and an isolating output buffer tapped near the middle of the chain. A nominal pre-layout SPICE control sits near 633 MHz. Arm A lets the flow place and route each oscillator with the surrounding logic. Arm B places 16 copies of one hardened oscillator macro on a regular grid. The logical circuit is identical in both arms; the physical implementation method is the experimental variable. Figure 1 summarizes the design.

One design, two layout methods. Nominal post-layout simulation isolates the implementation contribution; silicon adds variation.

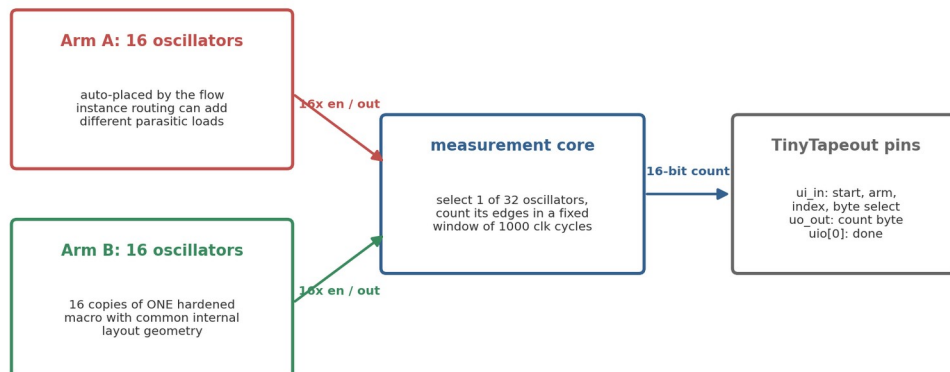


Figure 1. Block diagram of the two-arm design. Arm A lets the flow place and route each oscillator separately; Arm B repeats one hardened macro with a common internal layout.

One caveat matters for interpretation: the two arms are not identical apart from internal-layout matching. Hardening inserts input and output delay buffers at the Arm B macro boundary that Arm A does not have, the sixteen Arm B macros occupy a regular block on one side of the tile while Arm A fills the rest, and the two arms differ in local decap and power-delivery geometry. The comparison is therefore a matched hardened-macro implementation against a conventional automated standard-cell implementation, not two circuits that differ only in internal routing; the ring loops themselves stay logically equivalent. The main results below come from a coherent build of the current RTL that passes the physical checks (Magic and KLayout DRC, XOR, LVS, antenna, detailed route, power grid) with zero violations; two earlier builds, an archived dual-arm snapshot and a 32-oscillator layout, are reported for contrast (see SIGNOFF.md in the repository).

4. Method

The analysis is four separate operations, each traceable to a checked-in file. First, the physical flow produces the gate-level netlist, DEF, and nominal-corner SPEF. Second, a structural verifier (`verify_ring_topology.py`) parses the final netlist and confirms that every Arm A oscillator kept exactly one enable NAND, 30 inverters, and one tap buffer, with no cell inserted into the loop; the deck generation is only valid if this holds. Third, the generator reconstructs that verified topology from nominal SKY130 transistor-level cell models and attaches each ring net's SPEF *D_NET total capacitance to the matching node

as a grounded lumped capacitor. It does not simulate the extracted network itself. Fourth, ngspice transient simulation (1.8 V, 27 C, nominal TT models) measures each oscillator's frequency, with a parallel control deck carrying no extracted capacitance. The SPEF declares PIN_CAP NONE, so pin capacitance is not part of the transferred load.

On the chip, one oscillator runs at a time. The combined simulation deck enables all 16 reconstructed oscillators concurrently, which is equivalent under this model because they share only an ideal supply and the reduced model carries no coupling between them. Distributed wire resistance is omitted entirely; its effect is not quantified here and is one reason the distributed-RC comparison is required. Each oscillator starts disabled and is released by an enable pulse, which avoids a simulation artefact where an artificial initial condition excited an unintended ring mode. Frequency is measured over 20 periods after startup.

For the matched arm, the hardened macro is extracted and simulated once. The 16 instances share one internal GDS, so this single result serves as the internal-layout reference for all of them.

For each automatically placed array I report mean, standard deviation, range, peak-to-peak spread relative to the mean, and the Pearson correlation between frequency and extracted ring capacitance. Instances inside one routed design are not independent samples from a chip population, so these are descriptive statistics for that layout. Standard deviations are population values across those instances, which are the complete set for a layout, not a sample from a wider population. The extracted capacitance is also the only spread-producing input to the model, so a strong frequency-capacitance correlation is expected almost by construction. The coefficient confirms the model behaves; the spread it produces, and the slope, are the physically interesting parts.

5. Automatically placed arrays

5.1 Earlier 32-oscillator layout

The earlier build placed all 32 oscillators automatically. In the no-parasitic control every deck produced the same nominal frequency, approximately 633.640 MHz, which checks the generator itself. With extracted capacitance the mean was 567.6 MHz with a population standard deviation of 10.8 MHz (1.90%). Frequencies ranged from about 539 to 589 MHz, a 50.2 MHz or 8.8% peak-to-peak spread.

Extracted ring capacitance ranged from 7.4 to 17.8 fF. Frequency and capacitance had Pearson $r = -0.997$ with a fitted slope of about -4.93 MHz/fF (Figure 2a). Correlations with placement coordinates were much weaker ($|r| < 0.27$; Figure 2b). For this layout the spread behaves like an instance-specific routing load, not a smooth die-wide gradient.

Arm A: nominal lumped-C model predicts layout-dependent spread

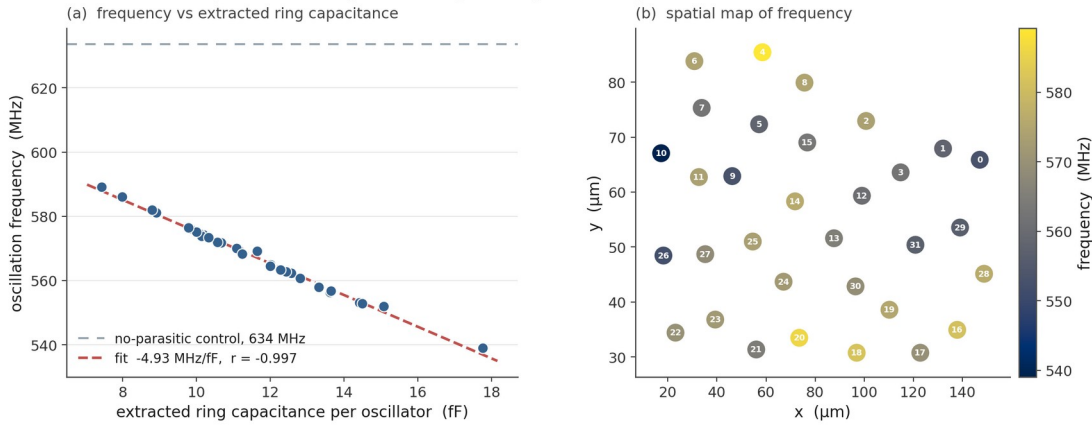


Figure 2. Nominal post-layout results for the earlier 32-oscillator layout: frequency versus extracted ring capacitance with the no-parasitic control, and a spatial frequency map.

5.2 Coherent dual-arm build

Arm A of the coherent dual-arm build contains 16 automatically placed oscillators. Their nominal post-layout frequencies have a mean of 551.7 MHz and a 58.0 MHz peak-to-peak range, 10.5% of the mean, with $r = -0.999$ against extracted ring capacitance (Figure 4). RO15 carries an unusually heavy routing load, 24.35 fF where the other fifteen span 11.7 to 16.95 fF, and sits at 508.5 MHz. Because peak-to-peak is outlier-sensitive, a leave-one-out check matters: without RO15 the spread is 4.60% peak to peak and the SD 1.25%, while the capacitance-frequency relation barely moves ($r = -0.9987$, rank correlation -0.996). The outlier is a real product of the router, not a measurement fault, so it stays in the primary result; the sensitivity numbers show exactly how much of the extreme range it carries (sensitivity.py recomputes all of this from the raw logs). This build passes Magic DRC, KLayout DRC, XOR, LVS, antenna, and power grid with zero violations, so the dispersion estimate and the candidate GDS come from the same run.

The three builds do not share a frequency pattern or a spread: 10.5% here, 5.4% in an earlier archived dual-arm snapshot, 8.8% in the 32-oscillator build. These are historical observations consistent with run-specific routing assignments; they are not controlled replicates, since the builds differ in source revision and settings as well as seed. The planned multi-seed study addresses that properly. A stronger cross-build check than comparing spread numbers is per-instance prediction: a linear capacitance fit trained only on the 32-oscillator build (624.6 MHz - 4.93 MHz/fF) predicts this build's 16 frequencies with MAE 0.77 MHz (0.14% of the mean), RMSE 1.18 MHz, and rank correlation 0.997, with the largest error on the outlier RO15 (-0.80%). The capacitive mechanism transfers between independent routing runs even though each pattern is unique.

6. Matched-macro arm

The matched construction hardens one oscillator as a 60 x 40 micrometre macro. The macro layout passes the available DRC, LVS, antenna, and connectivity checks, and Arm B places 16 instances of it on a regular grid.

The extracted macro carries about 11.0 fF of total ring capacitance. One nominal post-layout simulation gives 569.5 MHz against a no-parasitic control of 633.15 MHz. The earlier Arm A capacitance fit predicts 570.2 MHz at that load, 0.12% away, a useful cross-check on the model. The macro result also lands within 0.35% of the earlier Arm A mean, so matching did not move the operating point.

Figures 3 and 4 draw Arm B as a single horizontal reference line at 569.5 MHz, because the sixteen instances share one internal layout and there is only one extracted simulation behind it. By construction the internal layout contributes zero spread; fabricated Arm B instances will still differ through device mismatch, top-level routing, supply, and temperature. The simulation establishes only the internal-layout contribution under nominal device parameters. Total Arm B dispersion requires per-instance integration parasitics and fabricated-device measurements, and whether Arm B ends up with a smaller total spread than Arm A is the measurement the chip exists to make.

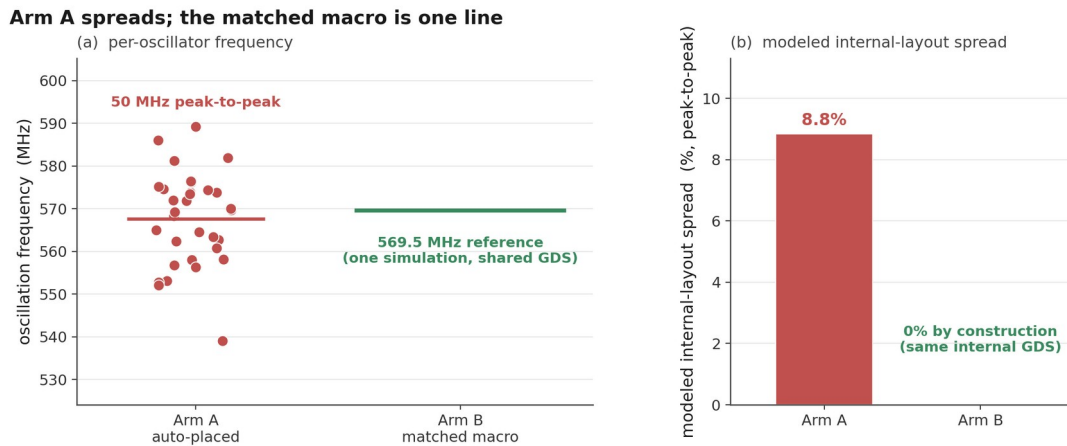


Figure 3. The earlier 32-oscillator array beside the matched-macro reference line at 569.5 MHz.

The dual-arm chip: Arm A spreads, Arm B is one layout

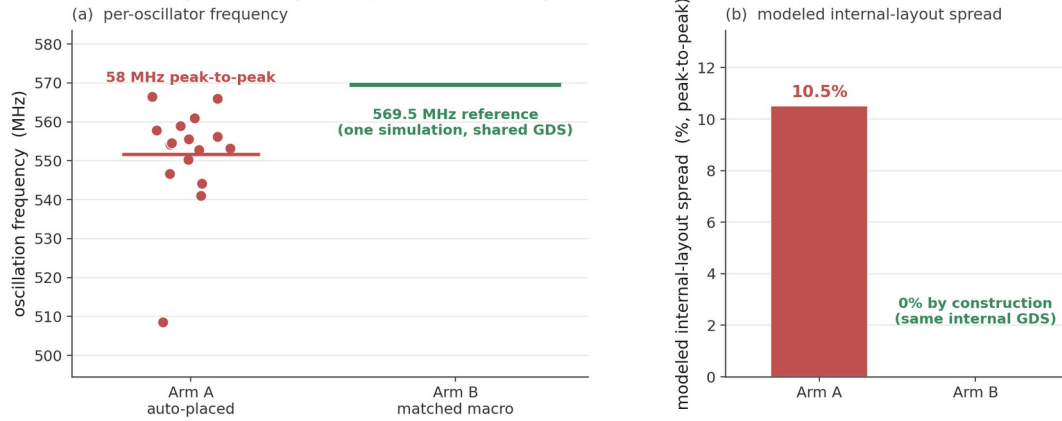


Figure 4. Arm A of the coherent dual-arm build (10.5% peak to peak) beside the matched-macro reference line.

7. Planned silicon test

Why could a repeatable mask-defined pattern matter to a PUF at all? A layout component shared across dies can reduce uniqueness or make some comparisons predictable, but prior work also shows systematic structure is not automatically exploitable: Wilde, Hiller, and Pehl found adjacent-oscillator comparisons suppressed the spatial structure in their data and their predictor could not beat its baseline [4]. Which way this design falls is a question for fabricated dies, not for the nominal model. A toy population model and a first-order mismatch-scale estimate live in the repository's supplementary material (`sim/montecarlo.py`, `sim/spice/mc/`); their outputs depend on assumed amplitudes and a $\sqrt{31}$ scaling that the PDK's global mismatch draw does not really support, so no number from them appears here.

The question for silicon is whether Arm A retains more of its nominal layout pattern across dies than Arm B. The threat model I have in mind is concrete: an attacker knows the public design and mask and holds measurements from other dies of the same design, but none from the target die, and asks whether the shared deterministic layout component lets them predict the target die's pair ordering above the relevant per-bit baseline. Predictor accuracy is then judged against that baseline with whole chips held out, not against 50%. Testing it needs multiple chip IDs, repeated measurements, matched voltage and temperature settings, and a fixed comparison rule; the firmware records chip and condition labels so groups stay separate. The planned metrics are repeatability within chip and condition, centered pattern correlation across chips under the same condition, inter-chip Hamming distance for a predeclared set of adjacent comparison pairs, and within-chip response changes across voltage and temperature. Results count only when the grouping and completeness requirements are met; a single chip or an incomplete oscillator vector does not support a population claim.

8. Limitations

This study is pre-silicon, and its model is deliberately simple. Nominal transistor models carry no random local mismatch. Lumped capacitance omits distributed RC and dynamic coupling; a distributed-RC extraction would test whether the frequency ordering and the spread survive a fuller electrical model, and running it on the fresh build is the obvious next check. The three builds reported here are not a controlled replicate set; they differ in source revision and tool settings as well as seed, so I do not aggregate their spreads into one distribution. A planned multi-seed study that varies only the place-and-route seed from one frozen source, config, and toolchain will give a proper spread distribution at essentially no cost next to fabrication. Instances within a single layout are also related observations, so instance-level confidence intervals would overstate the evidence. The Arm B reference is one simulation of one macro and cannot quantify fabricated Arm B variation. The 40-run global Monte Carlo cannot reproduce independent local device mismatch. Voltage, temperature, supply noise, ageing, package, and measurement-system effects are uncharacterized until chips exist, and uniqueness, reliability, min-entropy, and attack success all need a multi-chip data set with a stated threat model. The ripple counter is clocked by the oscillator itself, and its behaviour at the fastest corner is checked only in behavioural simulation; the standard-cell flop at the oscillator-to-counter boundary still needs extracted, across-corner timing before the fast end of the range can be trusted. None of this erases the modelled dispersion; it bounds what can be concluded from it.

9. Conclusion

Under a reduced lumped-capacitance model of the verified post-route topology, automated physical implementation gave the 16 oscillators of the coherent current build a 10.5% peak-to-peak nominal frequency dispersion (4.6% without the single heavy-routing outlier; SD 2.36% and 1.25% respectively). Two earlier uncontrolled builds showed 8.8% and 5.4% with different per-instance patterns, and a capacitance fit trained on one build predicts another's frequencies to 0.14% mean absolute error. A hardened macro provides a 569.5 MHz shared-internal-layout reference for Arm B. What this establishes is that the automated flow assigned substantially different routing capacitance to logically identical oscillators, and that under nominal device assumptions this predicts a frequency dispersion much larger than zero, with a mechanism that transfers across routing runs.

Uniqueness, reliability, min-entropy, and attack success are not evaluated pre-silicon. Whether the mask-defined pattern survives fabrication, and how it compares against real device mismatch, will be settled by measuring both arms of the fabricated chips under the protocol in the firmware.

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